Miguel Aenlle

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EDUCATION

M.S. in Computer Science, University of Illinois Urbana-Champaign *05/2028*

B.S. in Computer Science, University of Illinois Urbana-Champaign *05/2027*

- **GPA:** 4.0/4.0
- **Relevant Coursework:** Distributed Systems, Cloud Computing, Operating Systems, Systems Programming, Machine Learning Systems, Database Systems, Deep Learning, Computer Vision, Data Structures, Algorithms, Computer Architecture, C++, Java

EXPERIENCE

Software Engineer, **PrairieLearn** *08/2026 – Present*

- Architected agentic system automating course authoring, autograder debugging and analytics on PrairieLearn's code-based platform
- Isolated untrusted LLM-generated code in per-agent Linux VM sandboxes, deploying serverless agent fleet on Cloudflare Workers
- Sustained agent sessions past 10-minute sandbox TTLs by checkpointing state to R2, socket-streaming live progress to instructors

Software Engineer Intern, **Adobe** *05/2026 – 08/2026*

- Reduced manual ad campaign creation time to 5 minutes for Fortune 500 clients by architecting multi-agent generation workflow
- Built multimodal retrieval pipeline with transcription, Gemini, and VLM filtering over 10,000+ assets and Adobe templates per client
- Partnered with product managers on 30+ real-world eval scenarios and built harness measuring output correctness and latency

Software Engineer Intern, **PrairieLearn** *08/2025 – 05/2026*

- Led R&D of vision-LLM grading system achieving 99% accuracy on 10,000+ submissions; co-authored AIED 2026 paper on results
- Enabled long-running, resumable multi-agent rubric editing by building PostgreSQL-backed durable workflow infrastructure on EC2
- Shipped to 180,000+ students via concurrent batch inference, socket-streamed grading progress, and usage-based Stripe billing

Software Engineer Intern, **PrairieLearn** *12/2024 – 08/2025*

- Engineered image-capture API for 180,000+ students, ingesting assessment scans from webcams, phones, and document cameras
- Protected cross-device image uploads with CSRF tokens issued per session, blocking forged submissions to student assessments
- Rebuilt staff navigation system across 80+ pages with instructor feedback, gating pages and features by user access level

Software Engineer, **Normandy Remodeling** *08/2022 – 01/2025*

- Redeveloped legacy Microsoft Access database as full-stack web platform for 200,000 leads, 33,000 customers, and 9,900 projects
- Built authentication and role-based access control from scratch, scoping data entry, access, search, and analytics across 10+ staff roles

RESEARCH AND PROJECTS

ML Systems Research Intern, **UIUC SSAIL** *08/2026 – Present*

- Building agentic environment to optimize scheduling, routing, and autoscaling policies for NVIDIA Dynamo LLM inference clusters
- Developing autoresearch loops and RL finetuning infrastructure, testing policies in NVIDIA DynoSim at 1,200x real-time speed

ML Systems Engineer, **OptiTrain** *06/2026 – Present*

- Built t3.micro control plane orchestrating 8 EC2 GPU nodes, detecting failures via heartbeats and restoring them from checkpoints
- Chaos-tested a 17 hour GPT-2 124M pretraining run to completion, self-healing 31 injected node failures at 167s median recovery
- Created observability stack in Prometheus and Grafana tracking node liveness, training progress, and recovery events cluster-wide
- Built Go and Kubernetes serving engine, raising tokens per second with vLLM continuous batching, KV caching, and dtype casting

Research Software Developer, **Google Summer of Code — Emory University** *06/2025 – 02/2026*

- Expanded a limited, proprietary gastric MRI training set by 1,600% via custom-trained conditional diffusion and GAN models
- Raised UNet segmentation accuracy from 39% to 80% on held-out gastric scans via the expanded dataset and hyperparameter tuning

LEADERSHIP AND ACTIVITIES

Web Team Lead, **HackIllinois** *07/2025 – 03/2026*

- Led 4-engineer team to ship admin, logistics, and marketing systems for UIUC's flagship hackathon, garnering 500,000+ page views
- Shipped registration and logistics platform serving 2,000+ attendees and 50+ staff members across three-day flagship event

TECHNICAL SKILLS

Languages: Python, Go, C++, C, Java, TypeScript, JavaScript, SQL, Bash, Swift, HTML, CSS

Systems / Cloud: AWS (EC2, S3, Lambda), GCP (Vertex AI, Compute), Cloudflare (Workers, R2), Kubernetes, Docker, Linux, Prometheus

AI / ML: PyTorch, torch.distributed, DDP, NCCL, NVLink, CUDA, vLLM, LangGraph, Hugging Face, scikit-learn, OpenCV, NumPy, Pandas

Web / Data: React, Node.js, Express, REST, tRPC, WebSockets, SocketIO, PostgreSQL, Redis, MongoDB, CI/CD, Git, GitHub Actions